Directions for questions 1 to 10: Select the correct alternative from the given choices.

1. An amount of ₹2,400 is divided among A, B and C in the ratio 3 : 4 : 5. Find the share of A.
(1) ₹600 (2) ₹800 (3) ₹1,000
(4) ₹300 (5) ₹900

2. A sum of money is divided among P, Q, R and S in the ratio of 5 : 9 : 11 : 13. If the share of R is ₹2,856 more than that of P, then what is the total amount of money distributed among the four?
(1) ₹18,088 (2) ₹20,088 (3) ₹22,088
(4) ₹24,088 (5) ₹10,088

3. **(a)** If x : y = 4 : 3, then (3x + 2y) : (3x − 2y) is _____.
(1) 3 : 1 (2) 1 : 3 (3) 9 : 1
(4) 1 : 9 (5) 6 : 1

(b) If $2x = 3y = 4z$, then find the value of $\frac{2x+3y+4z}{3x+4y-2z}$?
(1) 3 : 4 (2) 7 : 9 (3) 4 : 3
(4) 9 : 7 (5) 6 : 7

4. If a : b = 2 : 3 and b : c = 5 : 7, find a : b : c.
(1) 10 : 15 : 21 (2) 10 : 21 : 15
(3) 9 : 12 : 14 (4) 12 : 7 : 18
(5) 10 : 21 : 14

5. If a : b = 1 : 2, b : c = 4 : 3 and c : d = 1 : 5, find a : d.
(1) 4 : 3 (2) 2 : 3
(3) 2 : 5 (4) 2 : 15
(5) 1 : 5

6. The ratio of two numbers is 4 : 5. If 15 is added to each of the two numbers, the ratio becomes 7 : 8. What is the sum of the two numbers?
(1) 39 (2) 42 (3) 45
(4) 30 (5) 27

7. The present age of Ramnath is equal to Simson's age six years ago. Nine years hence, the ratio of Ramnath's age to Simson's age will be 21 : 23. What is Simson's age now?
(1) 60 years (2) 54 years (3) 48 years
(4) 66 years (5) 72 years

8. The ratio of the present ages of X and Y is 7 : 6. The ratio of the age of X six years hence, to the age of Y four years ago is 3 : 2. What is the ratio of the age of X three hence years to the age of Y three years ago?
(1) 19 : 15 (2) 17 : 13 (3) 15 : 11
(4) 13 : 9 (5) 15 : 23

9. The incomes of P and Q are in the ratio 5 : 4 and their expenditures are in the ratio 3 : 2. If each one saves ₹1,000, then what is the income of P?
(1) ₹1,500 (2) ₹2,000 (3) ₹2,250
(4) ₹2,500 (5) ₹3,000

10. x varies inversely as the square of y. When y = 8, x is 1. Find x when y = 4.
(1) 3 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$
(4) 16 (5) 4

PRACTICE EXERCISE

Directions for questions 1 to 15: Select the correct alternative from the given choices.

***1.** The ratio of the number of boys to the number of girls of a school is 6 : 5. If 24 more girls join the school, then the ratio of the number of boys to the number of girls of the school will be 54 : 49. What is the total number of students of the school?
(1) 550 (2) 594
(3) 838 (4) 682
(5) 729

2. Last year, the ratio of the number of employees in company A to that of company B was 5 : 7. This year, the number of employees in A increased by 2,000 and that of B decreased by 2,000, and the ratio of the number of employees of A to that of B became 7 : 5. Find the number of employees in company A this year.
(1) 5,000 (2) 7,000 (3) 2,000
(4) 1,000 (5) 3,000

***4.** The ratio of Bhushan's age seven years ago to Archana's age five years ago is 4 : 5. The age of Bhushan three years hence is equal to the age of Archana two years ago. Find the age of Bhushan two years hence.

(1) 40 years (2) 37 years
(3) 35 years (4) 42 years
(5) 36 years

***5.** The ratio of the present age of Varun to that of his son is 13 : 4. After 15 years, Varun's age will be twice his son's age. If Varun's wife's age was 30 years when her son was born, then find the present age of Varun's wife.

(1) 42 years (2) 36 years
(3) 45 years (4) 12 years
(5) 56 years

***6.** The ratio of the present age of Asha to that of Bhanu is 6 : 5. If Vipul is two older than Asha, then what is the ratio of the present age of Vipul and Bhanu?

(1) 3 : 5 (2) 8 : 5
(3) 12 : 13 (4) Cannot be determined
(5) 5 : 8

***7.** The ratio of the present age of a mother and daughter is 13 : 4. Four years hence, the ratio of the ages of daughter and that of father will be 6 : 19. If the difference between the age of father and mother is eight years, then what is the present age of the father?

(1) 25 years (2) 34 years
(3) 41 years (4) 51 years
(5) 63 years

***8.** A certain sum was to be divided among Akash, Brijoy and Charan in the ratio of 14 : 17 : 19 but erroneously it was divided in the ratio of 8 : 9 : 13. As a result Charan got ₹344 more than what he was supposed to get. What was the total amount distributed among the three?

(1) ₹6,450 (2) ₹6,600
(3) ₹6,750 (4) ₹6,900
(5) ₹7,900

***10.** In a bag there are certain number of coins of the denominations ₹1, ₹2 and ₹5 in the ratio of 5 : 7 : 9. If the total value of the coins in the bag is ₹192, find the number of ₹1 coins in the bag.

(1) 5 (2) 10 (3) 15
(4) 20 (5) 25

11. In a class, the number of boys and girls are in the ratio 6 : 5. All the boys and girls took an exam. The ratio of the number of boys passed to that of boys failed is 2 : 1. The ratio of the number of girls passed to that failed is 3 : 2. If the total number of students failed in the examination is 28, then what is the total number of students in the class?

(1) 77 (2) 88 (3) 99
(4) 110 (5) 66

12. In a factory, the number of engineers, skilled workers and unskilled workers are in the ratio of 5 : 6 : 7. Their average salaries are in the ratio of 5 : 3 : 2. If the total wages paid to all the employees is ₹22,80,000, then what are the total wages paid to the skilled workers?

(1) ₹5.4 lakh (2) ₹6 lakh (3) ₹6.6 lakh
(4) ₹7.2 lakh (5) ₹9.6 lakh

13. Find the fourth proportion of the numbers 18, 6 and 9.

(1) 4 (2) 3 (3) 4.5
(4) 12 (5) 2.5

14. If x is subtracted from each of 17, 23 and 32, then the three numbers thus formed will be in continued proportion. What is the value of x?

(1) 3 (2) 6 (3) 5
(4) 4 (5) 2

15. A telephone bill consists of two parts. One part is fixed rental whereas the other part varies directly with the number of calls made. When the calls made are 60 and 120, the total amounts of two bills are ₹270 and ₹390 respectively. Find the bill amount for 180 calls.

(1) ₹470 (2) ₹510 (3) ₹590
(4) ₹660 (5) ₹720